

Medical Billing EHR/EMR Software

Architecture & Feature Guide

1. Core Modules

- Patient Management
- Patient registration & demographics
- Insurance eligibility verification (real-time)
- Patient portal (self-scheduling, messaging, documents)
- Consent forms & e-signatures

Electronic Health Records (EHR/EMR)

- Clinical notes (SOAP notes, progress notes)
- Problem lists, allergies, medications
- Lab/imaging orders & results
- Referral management
- Care plans & treatment history

Medical Billing Engine

- Charge capture from clinical encounters
- ICD-10 / CPT / HCPCS code mapping
- Claims generation (CMS-1500, UB-04)
- Clearinghouse submission (EDI 837)
- ERA/EOB posting (EDI 835)
- Denial management & appeals tracking
- Patient statements & payment plans

Scheduling

- Multi-provider, multi-location calendar
- Appointment reminders (SMS/email)
- Waitlist management
- Telehealth integration

Reporting & Analytics

- AR aging reports
- Collection rates, denial rates
- Payer mix analysis
- Provider productivity dashboards

2. Compliance & Standards (Non-Negotiable)

Standard-Purpose

1. HIPAA
2. PHI privacy & security

3. HL7 FHIR R4
4. Data interoperability
5. EDI X12 837/835/270/271
6. Claims & eligibility
7. ICD-10-CM/PCS
8. Diagnosis coding
9. CPT / HCPCS
10. Procedure coding
11. NCPDP
12. Pharmacy/e-prescribing
13. ONC Certification
14. Required for Medicare/Medicaid
15. SOC 2 Type II
16. Security audit

3. Technology Stack

Backend

Language:

Python (Django/FastAPI) or Java (Spring Boot) or Node.js

Database:

PostgreSQL (primary), Redis (caching), Elasticsearch (search)

Queue:

RabbitMQ or Kafka for claims processing

Storage: AWS S3 (encrypted) for documents/imaging

Frontend

- React or Angular (complex clinical forms benefit from Angular's form handling)
- Tailwind CSS or Material UI
- Real-time updates via WebSockets (for scheduling, alerts)
- Infrastructure
- AWS GovCloud or Azure Government (HIPAA BAA available)
- Kubernetes for container orchestration
- End-to-end encryption (AES-256 at rest, TLS 1.3 in transit)
- Audit logging (every data access must be logged)

Key Integrations

- Clearinghouses: Change Healthcare, Availity, Waystar
- Labs: LabCorp, Quest (HL7 interfaces)
- Pharmacies: Surescripts (e-prescribing)
- Payment processors: Stripe (HIPAA-eligible), InstaMed

- Telehealth: Zoom for Healthcare, Daily.co

4. Database Schema — Key Entities

patients → encounters → diagnoses (ICD-10) → procedures (CPT) → charges → claims → claim lines → claim submissions → payments / adjustments

providers → provider groups → locations

insurance payers → fee schedules → allowed amounts

5. Revenue Cycle Management (RCM) Workflow

Patient Check-in

- Eligibility Check (270/271)
- Encounter / Clinical Documentation
- Charge Capture
- Coding (ICD-10 + CPT)
- Claims Scrubbing (edit checks)
- EDI 837 Submission to Clearinghouse
- Payer Adjudication
- ERA 835 / EOB Posting
- Patient Balance Statement
- Collections / Write-offs

6. Build vs. Buy Decisions

Component

Use Existing Core EHR workflows

- ☒ Custom ICD-10/CPT code sets
- ☒ License from CMS/AMA

Clearinghouse

- ☒ Change Healthcare / Availity

E-Prescribing

- ☒ Surescripts API, Payment processing
- ☒ Stripe / InstaMed

Lab interfaces

✓ HL7 feeds from labs

Drug database

✓ Wolters Kluwer / FDB

7. Development Phases & Timeline

Phase 1: Patient registration, scheduling, basic EHR charting, user auth with RBAC

Phase 2 : Billing engine, claims submission, ERA posting, patient portal, eligibility checks

Phase 3: Advanced analytics, denial management, mobile apps, FHIR APIs, ONC certification

Phase 4: AI-assisted coding, clinical decision support, population health tools

8. Team Required

Healthcare IT Compliance specialists (HIPAA, ONC)

Backend engineers (billing/claims logic is complex)

Frontend engineers

DevOps/Security engineer

1 Medical coder/consultant (for CPT/ICD-10 rules)

1 QA engineer with healthcare testing experience